

Motion Detection App

Development plan following the 8-stage architecture

8 PHASES / ABOUT 1 WEEK / 1 TO 2 HOURS A DAY

01 Overview

The plan builds the app in the same order as the pipeline: each phase adds one stage, and every phase ends with a check you can run, so problems are found early. It assumes about 1 to 2 hours a day and takes roughly a week.

- The project zip already contains the code for phases 1 to 6. Use the **Done when** checks to verify each stage, or rebuild it phase by phase to learn how it works.

02 Timeline

Phase	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7+
0 Setup							
1 Webcam capture							
2 Preprocess, diff, threshold							
3 Rest or motion decision							
4 Direction							
5 Test and tune							
6 Polish and file management							
7 Upgrades (optional)							

Phases 0 and 1 share Day 1. The dashed bar is optional work after the core app is finished.

03 Phases

#	Phase	What to build	Module(s)	Done when
0	Setup	Project folder, virtual environment, requirements.txt, git repo	requirements.txt .gitignore	python -c "import cv2" works
1	Webcam capture	Open the camera, mirror the frame, handle "camera not found", quit on q	camera.py	Live video shows and closes cleanly
2	Preprocess, diff, threshold	Grayscale and blur, absdiff against the previous frame, black and white mask, changed-pixel count	preprocess.py detector.py	In the mask window only moving parts are white, and a still scene is almost black
3	Rest or motion decision	Threshold on the score, rest delay so the label does not flicker, on-screen label	state.py overlay.py config.py	Sitting still for 30 seconds never shows "Motion detected", and waving always does
4	Direction	Centroid of the mask, history of the last 6 centroids, dx/dy to Left/Right/Up/Down, dead zone	detector.py direction.py	10 slow swipes in each direction are read correctly at least 8 times
5	Test and tune	Unit tests with synthetic frames, manual test matrix, tune the settings	tests/ config.py	pytest passes and the manual test matrix passes
6	Polish and file management	Command-line options, optional CSV event log, README, clean folder structure	main.py event_log.py README.md	A friend can set it up from the README alone
7	Upgrades (optional)	See section 06	new modules	Each upgrade works behind a flag without breaking the base app

04 Manual test matrix

Run each case in Phase 5 and tick the result. The completed table is useful evidence if you present the project to your teacher.

ID	Test case	How to test	Expected result	Pass / Fail
T1	Still, close to camera	Sit still for about 30 seconds, roughly 0.5 m away	Body in rest the whole time	
T2	Still, far from camera	Sit still for about 30 seconds, roughly 2 m away	Body in rest the whole time	
T3	Slow movement - left	About 2 seconds per swipe, 10 swipes	Motion detected - Left in at least 8 of 10	
T4	Slow movement - right	About 2 seconds per swipe, 10 swipes	Motion detected - Right in at least 8 of 10	
T5	Slow movement - up	About 2 seconds per swipe, 10 swipes	Motion detected - Up in at least 8 of 10	
T6	Slow movement - down	About 2 seconds per swipe, 10 swipes	Motion detected - Down in at least 8 of 10	
T7	Fast movement - left	Move quickly, 10 swipes	Motion detected - Left	
T8	Fast movement - right	Move quickly, 10 swipes	Motion detected - Right	
T9	Fast movement - up	Move quickly, 10 swipes	Motion detected - Up	
T10	Fast movement - down	Move quickly, 10 swipes	Motion detected - Down	
T11	Normal room light	Alternate between sitting still and moving	Correct label each time	
T12	Dim light	Alternate between sitting still and moving	Correct label; if false motion appears, raise blur or threshold	
T13	Lamp switched on or off	Sit still while someone switches a lamp	Ideally stays "Body in rest"; a brief flash is tolerable	
T14	Someone walking in the background	Sit still while another person walks behind you	Motion detected (reacts to any movement)	
T15	Hand movement only	Keep the body still and swipe one hand	Motion detected with correct direction	
T16	Whole-body movement	Lean or step across the frame	Motion detected with correct direction	

05 Risks and fixes

Risk	What to do	Setting or option
Light flicker or auto-exposure causes false motion	Raise the blur kernel or the threshold	blur_kernel, motion_threshold
Camera will not open on Windows	Try another camera index and close apps like Zoom or Teams	--camera 1
Direction is wrong when an object enters or leaves the frame	Raise the minimum shift or the history length	min_shift, history_length
Label flickers between rest and motion	Raise the number of still frames required before rest	rest_delay_frames
Detects movement that is not a person	Add person detection in Phase 7	Upgrade 1

06 Phase 7 upgrade ideas

#	Upgrade	What it adds
1	Person-only detection	MediaPipe Pose or YOLO, so a pet or a curtain does not trigger it
2	Optical flow	calcOpticalFlowFarneback for more accurate direction, including diagonals
3	Speed estimate	Speed from the centroid shift per second
4	Snapshot or alert	Save the frame or play a sound when motion starts
5	Simple GUI	Sliders for the threshold and shift, so tuning does not need a restart